

International Islamic University Chittagong
Department of Computer Science and Engineering
B. Sc. in CSE Mid-term Examination, Spring 2024
Course Code: CSE 2423 Course Title: Database Management System
Total marks: 30 Time: 1 hour 30 minutes

[Answer all the questions. Figures in the right-hand margin indicate full marks.]

- | | | CO | DL |
|--|---|---------|----|
| 1. | | | |
| a) | Define Database Management System. Write about any two of the real-life applications of Database Management System. | 2 · CO1 | R |
| b) | Describe about relational and object-oriented data model. | 3 CO1 | U |
| c) | Explain with example, how database management system resolve the problems of redundancy and atomicity | 5 CO2 | Ap |
| 2. | | | |
| a) | Consider the following specifications for designing a travel planning platform:
- Each user should have a unique user ID and associated profile details.
- Users can create multiple itineraries, each associated with a particular user.
- Each lists should include various destinations, and destinations may appear in multiple itineraries.
- Users can review accommodations, transportation, and attractions, each having associated details and ratings.
- The social component allows users to connect with friends, share itineraries, and view each other's travel experiences. | 5 CO1 | Ap |
| Design the E-R diagram for the above <i>problem</i> . Consider all the necessary mapping cardinalities, attributes and participation constraints. Your database should store information the relationships between users, lists, destinations, reviews, and social interactions and encode the following information. | | | |
| b) | Construct an ER-Diagram of 3 levels for Vehicles such as Bus, Truck and Rickshaw using specialization and generalization hierarchy system. Justify your placement of attributes at each level of the hierarchy. | 3 CO1 | Ap |
| c) | Write down the problems of using Cartesian Product. How could it be resolved? Explain with example. | 2 CO2 | U |
| OR | | | |
| Why should we avoid natural join? Explain with example | | | |
| 3. | | | |
| a) | Write names of 2 aggregate functions with their syntax in relational algebra. | 2 CO1 | R |
| b) | classroom (<u>building</u> , <u>room number</u> , capacity)
department (<u>dept name</u> , building, budget)
course (<u>course id</u> , title, dept_name, credits)
instructor (<u>ID</u> , name, dept_name, salary)
section (<u>course id</u> , <u>secid</u> , <u>semester</u> , <u>year</u> , building, room number, time slot id) | 8 CO1 | Ap |

teaches (ID, course_id, sec_id, semester, year)
student (ID, name, dept_name, tot_cred)
takes (ID, course_id, sec_id, semester, year, grade)

Consider the relational database of a university database management system, where the primary keys are underlined. Write queries in relational algebra to answer the following questions -

1. Find the ID, name and salary of each instructor in the Physics department or those having salary greater than \$40,000
2. Find the ID and name of each instructor in a department located in the building 'Watson'
3. Find all the courses taught in the Fall 2023 semester but not in Spring 2024 semester.
4. Find the ID and name of those instructors who earn more than the instructors in the Physics department.

OR,

1. Find all instructor whose salary is above average for their department
2. Find all courses taught in both the Fall 2022 semester and in the Spring 2022 semester
3. Find all students who have taken all courses taught by instructor named 'John Doe'
4. List all departments along with number of students in each department.

- | | | Mark | CO | DL |
|--------|--|-----------|-----|----|
| Q1. a) | Define Database management system? What is data Abstraction? Write down the benefits of data abstraction | 1+2
=3 | CO1 | C1 |
| b) | How does database system ensures atomicity and concurrency? Explain each with example | 3 | CO1 | C2 |
| c) | Compare between DML and DDL. Write down the functions of different component of storage manager. | 2+2
=4 | CO1 | C1 |
| Q2. a) | How does a query being executed through different components of DBMS? Explain. | 3 | CO1 | C2 |
| b) | Consider a scenario where a student can register for courses. All student are enrolled to a specific department. Courses are taught by the teacher of the department which a student is enrolled. A teacher can be a course teacher also an advisor at the same time. Now draw a simple E-R diagram from the above scenario. Consider the necessary mapping cardinalities, participation constraints and attributes. | 3 | CO3 | C6 |

OR

- | | | | | |
|---|---|-----------|-----|----|
| b) | Construct an ER of the database used by the ABC super store, where persons with ID, name, age considered as either Customer or Employee. Customers may be either wholesale or retail customers. Employees could be either operational or manager. The manager could control the employees to ensure proper sell of the products to the customers. Consider the necessary mapping cardinalities, participation constraints and attributes. | 4 | CO3 | C6 |
| c) | What do you understand by attributes? Write down the types of attributes. Explain at least three of them with proper example. | 3 | CO1 | C1 |
| Q3. a) | Write down the fundamental operations of the relational algebra and explain them with proper example. | 4 | CO1 | C2 |
| b) | passenger (pid, pname, pgender, party)
flight (flg, fdate, time, src, dest)
book (pid, fid) | 3x2
=6 | CO2 | C3 |
| Consider the relational database given above, where the primary keys are underlined. Give an expression in SQL. | | | | |

International Islamic University Chittagong

Department of Computer Science and Engineering

Mid-Term Examination, Spring-2023

Semester: 4th

Course Code: CSE-2423

Course Title: Database Management System

Time: 1 Hour 30 minutes

Marks: 30

Mark CO DL

relational algebra:

1. Get the details of flights that are scheduled on both dates 01/03/2023 and 03/03/2023 at 18:00 hours.
2. Get the details of flights that are scheduled on either of the dates 05/03/2023 or 07/03/2023 and booked for passenger 'P01'
3. Find the passenger who have booked same flight as 'P02'

OR

b) **Product**(P-id, P-name, Unit_price)

Sales(S-id, P-id, C-id, Emp_id, Data_of_Sale, Time, Quantity)

Customer(C-id, C-name, Mobile, Street, City, Country)

Employee(Emp-id, E-name, Designation, Commission, Salary, Mobile, City, Country)

3x2 CO2 C3

=6

Consider the relational database given above, where the primary keys are underlined. Give an expression in the relational algebra:

- 1) Find those Employees and their sales details who have sold more than 5000 on '20-Nov-2017'
- 2) Find the customers details who have served by employee 'X'
- 3) Find the customers who have purchased same products as 'Y' before '20-Nov-2017'.

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B. Sc. in CSE Mid-term Examination, Autumn 2023
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[Answer all the questions. Figures in the right-hand margin indicate full marks.]

- | | | | |
|----|--|----|-------|
| 1. | | CO | DL |
| a) | Describe different types of DBMS users. | 2 | CO1 R |
| b) | Imagine you're developing a healthcare management system for a hospital. Different users like doctors, nurses, and administrative staff need varying levels of access to patient records. How can provide users with an <i>abstract view of the data</i> in the database system? Explain with example. | 3 | CO1 U |

Or,

What is **data model**? Briefly describe 2 different types of data model with examples.

- | | | | |
|----|---|---|--------|
| c) | Consider a hotel reservation system for a chain of hotels. The system can handle room availability, guest reservations, and payment processing. | 5 | CO2 Ap |
|----|---|---|--------|

Issue-1: A guest successfully reserves a room but experiences a system failure prior to receiving a confirmation and before the room availability is properly updated.

Issue-2: Imagine two guests trying to book the same room at the same time.

Issue-3: A malicious user attempts to alter reservation records or manipulate room availability data.

How do the above issues **relate to the problems faced by file systems** while storing information? Explain with example how these can be resolved using database.

- | | | | |
|----|---|---|--------|
| 2. | | 5 | CO1 Ap |
| a) | Suppose you are running a startup company entitled as " LIKHO " for a blog writing platform allowing users to post blogs, and to comment on other users' blogs. | | |
| | Design the E-R diagram for your company's data. Consider all the necessary mapping cardinalities, attributes and participation constraints. Your database should store information about users, bloggers, and blogs, and encode the following information: | | |

- Every user has a user ID and a name.
- Every blog has a blog ID, a text content and one author, who is a blogger.
- Every blogger is a user.
- Every blogger also has a rating attribute.
- Users may comment on blogs.
- Every comment has an optional rating.

Or,

Construct an **E-R diagram** for a Grocery shop with a set of customers and a set of products. Customer is of two types such as retail and wholesale. Associate with each customer a log of the various products and providing each customer with the voucher, showing Total Amount being purchased. Consider the necessary mapping cardinalities, participation constraints and attributes.

- b) Suppose you are designing an ER diagram for a Library Management System. The system manages different types of library items, including Books and DVDs. 3 CO1 Ap

1) Now design a **specialization and generalization hierarchy**. Justify your placement of attributes at each level of the hierarchy.

2) Is your placement total or partial?

- c) Write down the problems of using **Cartesian Product**. How could it be resolved? Explain with example. 2 CO2 U

3.

- a) What is **Relational Algebra**? Write the difference between **inner join** and **outer join**. 2 CO1 R

Or,

State the names of **basic operators of Relational Algebra** with symbols and meaning.

- b) **Customer** (customer_id, customer_name, address, city, state, phone_number) 8 CO1 Ap
Store (store_id, store_name, store_address, store_phone)
Food (food_id, store_id, food_name, food_image, food_category, food_price)
Order (order_id, customer_id, store_id, food_id, shipment_address, payment_method, quantity, total_price, order_date)

Consider the relational database of an online food app, where the primary keys are underlined. Write **queries in relational algebra** to answer the following questions -

1. Find the names of customers who has ordered between 1000 Tk and 2000 Tk.
2. Find the food items with their prices from a specific store. (store_name = "Well Food")
3. Find the total number of food items available in each store.
4. Find the order details of those customers who lives in the same city where the customer "X" lives.

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[Answer all the questions; in some questions, there are options;
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CO DL

1. a. Describe database instance and schema with example? What are the functions of a Storage Manager? 3 CO1 Un
- b. How does database system ensures atomicity and concurrency? Explain each with example. 4 CO1 Un
- OR
Compare among Primary key, Super key, Candidate key, and Foreign key with examples.
- c. What is Data Model? Give examples of some popular data model. Compare between DDL and DML. 3 CO1 Re
2. a. Define cardinality ratio, partial, and total participation. 3 CO1 Re
- b. Construct an E-R diagram for a Grocery shop with a set of customers and a set of products. Associate with each customer a log of the various products and provided each customer with the voucher, showing Total Amount being purchased. An employee can sell any number of products and can also be a manager. Consider the necessary mapping cardinalities, participation constraints and attributes. 4 CO3 Cr
- c. Briefly explain different types of attributes with example. 3 CO1 Un
3. a. Consider the Following Relational Database of *Pure Islamic Bank*: 6 CO2 An
branch (branch_name, branch_city, assets)
customer (customer_name, customer_street, customer_city)
account (account_number, branch_name, balance)
loan (loan_number, branch_name, amount)
depositor (customer_name, account_number)
borrower (customer_name, loan_number)
Write Relational Algebra Expression (Any three)
(i) Find all customers who have a loan, an account, or both.
(ii) Find all customers who have a loan at the *Kumira* branch.
(iii) Find the names of all customers who have a loan in *GEC* branch but do not have an account in the bank.
(iv) Find the account numbers of the Millionaires of the bank.
- b. Describe the basic Operators of Relational Algebra with example. 4 CO2 Un
- Or
Explain how you could map ER diagram to database with example.

Answer all the questions

1. **A database needs to be developed that supports a supermarket.**

7

Design an entity-relationship diagram that captures, as far as possible, the requirements stated below. (Statements in brackets are additional explanations and not part of the requirements proper.) If you make any assumptions in your design, please write them down. Assumptions, however, must not contradict the requirements.

1. The supermarket sells products. Each product is uniquely identified by its product number. Moreover, a product has price. The supermarket also records how many items of each product are currently on stock.
2. A product can be on special offer. A special offer is valid for a period, which has a start date and an end date. A special offer refers to exactly one product. Two special offers for the same product can not have overlapping periods. A special offer for a product specifies that a certain number of items of that product are sold at a certain price (e.g., 2 items for 1,99 EUR).
3. There is a card scheme for which customers can register. A registered customer holds a customer card. Each registered customer has a customer number, which is unique. For each registered customer the supermarket records the name and the address of the customer and the day when the customer joined the card scheme.
4. When a customer buys one or more items and pays for them, a purchase is being completed. A purchase is uniquely identified by the number of the checkout where the customer pays, together with the date and the time when the purchase is made.
5. Some purchases are made by registered customers.
6. A purchase consists of one or more line items. Within a given purchase, each line item is uniquely identified by its line number. Each line item refers to a unique product. (In fact, the line items correspond to the items of the purchase). Some line items have an additional comment, which is a string. (The comment may say that a special price applies to the product item because it is part of a special offer.)

2. UPS prides itself on having up-to-date information on the processing and current location of each shipped item. To do this, UPS relies on a company-wide information system. Shipped items are the heart of the UPS product tracking information system. Shipped items can be characterized by item number (unique), weight, dimensions, insurance amount, destination, and final delivery date. Shipped items are received into the UPS system at a single retail center. Retail centers are characterized by their type, uniqueID, and address. Shipped items make their way to their destination via one or more standard UPS transportation events (i.e., flights, truck deliveries). These transportation events are characterized by a unique scheduleNumber, a type (e.g, flight, truck), and a deliveryRoute. 7

a. Please create an **Entity Relationship diagram** that captures this information about the UPS system. Be certain to indicate identifiers and cardinality constraints.

b. Please convert the ER diagram into a relational database schema. Be certain to indicate primary keys and referential integrity constraints.

3. **Customer Database** 7

Use the following database:

Customer(cid: integer, cname: string, rating: integer, salary: real)

Item(iid: integer, iname: string, type: string)

Order(cid: integer, iid: integer, day:date, qty:real)

Write the following Queries using Relational Algebra:-

- Find names of customers who've ordered item with id 100
- Find names of Customers who've ordered a laptop (i.e., an item of type "laptop")
- Find customers who've ordered a laptop and a desktop computer
- Find the names of customers who've ordered all item